



9969 - EXTREME: EXploiting TRansients of Extremely Magnified Events

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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JWST Proposal 9969 (Created: Friday, May 29, 2026, 9:00:14AM Eastern Standard Time) - Overview

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OBSERVATIONS

Folder	Observation	Label	Observing Template	Science Target
Observation Folder				
	1	MACS0416-24	NIRCam Imaging	(1) MACSJ0416.1-2403
	2	MACS0416-24	NIRCam Imaging	(1) MACSJ0416.1-2403
	3	MACS0416-24	NIRCam Imaging	(1) MACSJ0416.1-2403
	4	MACS0416-24	NIRCam Imaging	(1) MACSJ0416.1-2403
	5	MACS0416-24	NIRCam Imaging	(1) MACSJ0416.1-2403
	6	MACS0416-24	NIRCam Imaging	(1) MACSJ0416.1-2403

Folder	Observation	Label	Observing Template	Science Target
	7	MACS0416-24	NIRCam Imaging	(1) MACSJ0416.1-2403
	8	MACS0416-24	NIRCam Imaging	(1) MACSJ0416.1-2403

ABSTRACT

Direct observations of individual luminous stars at cosmological distances are possible with JWST thanks to its superior capability to detect stars which are magnified by factors of several thousand near the critical curves (CC) of galaxy clusters. The magnification near a CC is constantly modified by microlenses from the intracluster medium, and also from small scale perturbations in the distribution of dark matter (DM). These modify the observed flux as a function of time, ntroducing recognizable patterns in the distribution of observed microlensing events.

Recent NIRCam observations of the rich cluster MACS J0416.1-2403 (MACS0416) have revealed a record 3 microlensing events per epoch in this field, all from two galaxies at $z=1$ which are intersecting the CC. Repeated observations of this field will significantly increase the number of microlensing events, offering the best opportunity to study the most massive stars in these two galaxies, and provide new clues about DM. About 1 Cepheid is also expected to be detected in each of these two galaxies at $z=1$.

We propose a 10-day-cadence NIRCam monitoring on MACS0416 spanning approx. 3 months, to increase the number of detections and study three important issues:

- 1) Measure the tail of the stellar luminosity function at $z = 1$ through direct observation of the most luminous stars at $z=1$ and establish the validity of the Humphrey-Davidson limit for RSG stars
- 2) Search for the first Cepheid stars at $z=1$
- 3) Test dark matter models that challenge CDM (primordial black holes and wave DM), via the small scale lensing perturbations affecting the number and spatial distribution of observed stars.

OBSERVING DESCRIPTION

This program will execute a NIRCam time-domain imaging survey of the lensing cluster MACS0416-24.

The observations comprise 8 epochs at an even 10 ± 1 day cadence (some flexibility of ± 1 day regarding the scheduling as described in the

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The full set of observations spans the entire contiguous scheduling window of MACS0416-24 from late 2026 to early 2027, so the absolute timing constraints of each epoch are crucial to the overall program.

Our rapid cadence (10+/-1 days) results in high overlap of NIRCcam FOVs between successive search epochs, without the need to additionally constrain PA_V3.

We have purposely minimized the layout and timing constraints, to maximize scheduling flexibility within the full visibility window of the target field.

We use the NIRCcam filters F150W, F200W in the SW, and F277W, F356W in the LW.

The targeting (and filter choices) is similar to that of GO-1176, so as to maximize areal (and frequency) overlap with those early epochs.

One NIRCcam module is centered on the cluster, while the other module rotates around the cluster from epoch to epoch, with large adjacent-epoch overlap, and overlapping also existing JWST data from previous cycles. The centre of module B is chosen to maximize overlap with previous observations from cycle 1 (GTO program, PI : R. Windhorst).

Proposal 9969 - Targets - EXTREME: EXploiting TRansients of Extremely Magnified Events

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	MACSJ0416.1-2403	RA: 04 16 9.2870 (64.0386958d) Dec: -24 04 3.20 (-24.06756d) Equinox: J2000		
	<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>Coordinates updated to match HFF pointings as published in Lotz et al. (2017).</i> <i>Category=Clusters of Galaxies</i> <i>Description=[Rich clusters]</i>				

Proposal 9969 - Observation 1 - EXTREME: EXploiting TRansients of Extremely Magnified Events

Observation	Proposal 9969, Observation 1: MACS0416-24										Fri May 29 14:00:14 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRCcam Imaging										
	Comments: aligned to HFF WFC3/IR (allowing 90 deg rotations) epoch 1 / 3 enabling search for transients PA can vary while keeping module B on target										
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	MACSJ0416.1-2403	RA: 04 16 9.2870 (64.0386958d) Dec: -24 04 3.20 (-24.06756d) Equinox: J2000								
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Coordinates updated to match HFF pointings as published in Lotz et al. (2017). Category=Clusters of Galaxies Description=[Rich clusters]										
Template	Module		Subarray				Target Placement				
	ALL		FULL				Detector B1 (compact source)				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	INTRAMODULEX		4		STANDARD				2	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F150W	F277W	MEDIUM8	8	1	8	8	6699.744	270634	
	2	F200W	F356W	MEDIUM8	8	1	8	8	6699.744	270634	
Special Requirements	Between Dates 29-NOV-2026:00:00:00 and 15-DEC-2026:00:00:00 Aperture PA Range 0.34481904 to 13.34481904 Degrees (V3 0.0 to 13.0) Fiducial Point Override NRCB1_FULL 2 After 1 by 9 Days to 11 Days 3 After 1 by 19 Days to 21 Days 4 After 1 by 29 Days to 31 Days 5 After 1 by 39 Days to 41 Days 6 After 1 by 49 Days to 51 Days 7 After 1 by 59 Days to 61 Days 8 After 1 by 69 Days to 71 Days										

Proposal 9969 - Observation 2 - EXTREME: EXploiting TRansients of Extremely Magnified Events

Observation	Proposal 9969, Observation 2: MACS0416-24										Fri May 29 14:00:14 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
	Comments: aligned to HFF WFC3/IR (allowing 90 deg rotations) epoch 1 / 3 enabling search for transients										
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	MACSJ0416.1-2403	RA: 04 16 9.2870 (64.0386958d) Dec: -24 04 3.20 (-24.06756d) Equinox: J2000								
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Coordinates updated to match HFF pointings as published in Lotz et al. (2017). Category=Clusters of Galaxies Description=[Rich clusters]										
Template	Module		Subarray				Target Placement				
	ALL		FULL				Detector B1 (compact source)				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	INTRAMODULEX		4		STANDARD				2	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F150W	F277W	MEDIUM8	8	1	8	8	6699.744		
	2	F200W	F356W	MEDIUM8	8	1	8	8	6699.744		
Special Requirements	Aperture PA Range 13.34481904 to 26.34481904 Degrees (V3 13.0 to 26.0) Fiducial Point Override NRCB1_FULL 2 After 1 by 9 Days to 11 Days										

Proposal 9969 - Observation 3 - EXTREME: EXploiting TRansients of Extremely Magnified Events

Observation	Proposal 9969, Observation 3: MACS0416-24										Fri May 29 14:00:14 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
	Comments: aligned to HFF WFC3/IR (allowing 90 deg rotations) epoch 1 / 3 enabling search for transients										
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous	
	(1)	MACSJ0416.1-2403	RA: 04 16 9.2870 (64.0386958d) Dec: -24 04 3.20 (-24.06756d) Equinox: J2000								
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Coordinates updated to match HFF pointings as published in Lotz et al. (2017). Category=Clusters of Galaxies Description=[Rich clusters]										
Template	Module				Subarray				Target Placement		
	ALL				FULL				Detector B1 (compact source)		
Dithers	#	Primary Dither Type			Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions
	1	INTRAMODULEX			4		STANDARD				2
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F150W	F277W	MEDIUM8	8	1	8	8	6699.744		
	2	F200W	F356W	MEDIUM8	8	1	8	8	6699.744		
Special Requirements	Aperture PA Range 26.34481904 to 40.34481904 Degrees (V3 26.0 to 40.0) Fiducial Point Override NRCB1_FULL 3 After 1 by 19 Days to 21 Days										

Proposal 9969 - Observation 4 - EXTREME: EXploiting TRansients of Extremely Magnified Events

Observation	Proposal 9969, Observation 4: MACS0416-24									Fri May 29 14:00:14 GMT 2026
	Diagnostic Status: Warning									
	Observing Template: NIRCam Imaging									
	Comments: aligned to HFF WFC3/IR (allowing 90 deg rotations) epoch 1 / 3 enabling search for transients									
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous	
	(1)	MACSJ0416.1-2403	RA: 04 16 9.2870 (64.0386958d) Dec: -24 04 3.20 (-24.06756d) Equinox: J2000							
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Coordinates updated to match HFF pointings as published in Lotz et al. (2017). Category=Clusters of Galaxies Description=[Rich clusters]									
Template	Module		Subarray				Target Placement			
	ALL		FULL				Detector B1 (compact source)			
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions
	1	INTRAMODULEX		4		STANDARD				2
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
	1	F150W	F277W	MEDIUM8	8	1	8	8	6699.744	
	2	F200W	F356W	MEDIUM8	8	1	8	8	6699.744	
Special Requirements	Aperture PA Range 35.34481904 to 55.34481904 Degrees (V3 35.0 to 55.0) Fiducial Point Override NRCB1_FULL									
	4 After 1 by 29 Days to 31 Days									

Proposal 9969 - Observation 5 - EXTREME: EXploiting TRansients of Extremely Magnified Events

Observation	Proposal 9969, Observation 5: MACS0416-24									Fri May 29 14:00:14 GMT 2026
	Diagnostic Status: Warning									
	Observing Template: NIRCcam Imaging									
	Comments: aligned to HFF WFC3/IR (allowing 90 deg rotations) epoch 1 / 3 enabling search for transients									
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous	
	(1)	MACSJ0416.1-2403	RA: 04 16 9.2870 (64.0386958d) Dec: -24 04 3.20 (-24.06756d) Equinox: J2000							
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Coordinates updated to match HFF pointings as published in Lotz et al. (2017). Category=Clusters of Galaxies Description=[Rich clusters]									
Template	Module		Subarray				Target Placement			
	ALL		FULL				Detector B1 (compact source)			
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions
	1	INTRAMODULEX		4		STANDARD				2
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
	1	F150W	F277W	MEDIUM8	8	1	8	8	6699.744	
	2	F200W	F356W	MEDIUM8	8	1	8	8	6699.744	
Special Requirements	Aperture PA Range 44.34481904 to 65.34481904 Degrees (V3 44.0 to 65.0) Fiducial Point Override NRCB1_FULL									
	5 After 1 by 39 Days to 41 Days									

Proposal 9969 - Observation 6 - EXTREME: EXploiting TRansients of Extremely Magnified Events

Observation	Proposal 9969, Observation 6: MACS0416-24									Fri May 29 14:00:14 GMT 2026
	Diagnostic Status: Warning									
	Observing Template: NIRCam Imaging									
	Comments: aligned to HFF WFC3/IR (allowing 90 deg rotations) epoch 1 / 3 enabling search for transients									
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous	
	(1)	MACSJ0416.1-2403	RA: 04 16 9.2870 (64.0386958d) Dec: -24 04 3.20 (-24.06756d) Equinox: J2000							
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Coordinates updated to match HFF pointings as published in Lotz et al. (2017). Category=Clusters of Galaxies Description=[Rich clusters]									
Template	Module		Subarray				Target Placement			
	ALL		FULL				Detector B1 (compact source)			
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions
	1	INTRAMODULEX		4		STANDARD				2
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
	1	F150W	F277W	MEDIUM8	8	1	8	8	6699.744	
	2	F200W	F356W	MEDIUM8	8	1	8	8	6699.744	
Special Requirements	Aperture PA Range 53.34481904 to 73.34481904 Degrees (V3 53.0 to 73.0) Fiducial Point Override NRCB1_FULL 6 After 1 by 49 Days to 51 Days									

Proposal 9969 - Observation 7 - EXTREME: EXploiting TRansients of Extremely Magnified Events

Observation	Proposal 9969, Observation 7: MACS0416-24										Fri May 29 14:00:14 GMT 2026	
	Diagnostic Status: Warning											
	Observing Template: NIRCam Imaging											
	Comments: aligned to HFF WFC3/IR (allowing 90 deg rotations) epoch 1 / 3 enabling search for transients											
Diagnostics	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	MACSJ0416.1-2403	RA: 04 16 9.2870 (64.0386958d) Dec: -24 04 3.20 (-24.06756d) Equinox: J2000									
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Coordinates updated to match HFF pointings as published in Lotz et al. (2017). Category=Clusters of Galaxies Description=[Rich clusters]											
Template	Module		Subarray					Target Placement				
	ALL		FULL					Detector B1 (compact source)				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions		
	1	INTRAMODULEX		4		STANDARD				2		
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID		
	1	F150W	F277W	MEDIUM8	8	1	8	8	6699.744			
	2	F200W	F356W	MEDIUM8	8	1	8	8	6699.744			
Special Requirements	Aperture PA Range 62.34481904 to 82.34481904 Degrees (V3 62.0 to 82.0) Fiducial Point Override NRCB1_FULL 7 After 1 by 59 Days to 61 Days											

Proposal 9969 - Observation 8 - EXTREME: EXploiting TRansients of Extremely Magnified Events

Observation	Proposal 9969, Observation 8: MACS0416-24										Fri May 29 14:00:14 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRCcam Imaging										
	Comments: aligned to HFF WFC3/IR (allowing 90 deg rotations) epoch 1 / 3 enabling search for transients										
Diagnostics	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous	
	(1)	MACSJ0416.1-2403	RA: 04 16 9.2870 (64.0386958d) Dec: -24 04 3.20 (-24.06756d) Equinox: J2000								
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Coordinates updated to match HFF pointings as published in Lotz et al. (2017). Category=Clusters of Galaxies Description=[Rich clusters]										
Template	Module		Subarray					Target Placement			
	ALL		FULL					Detector B1 (compact source)			
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	INTRAMODULEX		4		STANDARD				2	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F150W	F277W	MEDIUM8	8	1	8	8	6699.744		
	2	F200W	F356W	MEDIUM8	8	1	8	8	6699.744		
Special Requirements	Aperture PA Range 71.34481904 to 91.34481904 Degrees (V3 71.0 to 91.0) Fiducial Point Override NRCB1_FULL 8 After 1 by 69 Days to 71 Days										